XAI Based Detection of Brain Tumors using MRI Images

A PROJECT REPORT

Submitted By

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BONAFIDE CERTIFICATE

Certified that this project report **XAI Based Detection of Brain Tumors using MRI Images** is the Bonafide work of Anwesha Goswami, Jyotirmay Shrestha, Sayantika Bhunia and Sneha Lahiri, who carried out the project

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TABLE OF CONTENTS

Chapter No.	Title	Page No.
1	Introduction	4
2	Motivation	5
3	Objectives	5
4	Methodology	6
5	Future Work	20
6	Literature Survey	20
7	Conclusion	21
8	References	22

Abstract

This paper presents a comprehensive study on automatic brain tumor segmentation and classification using Magnetic Resonance Imaging (MRI). Accurate tumor segmentation is essential for diagnosis, treatment planning, and prognosis, yet manual annotation of MRI scans is time-consuming and prone to inter-observer variability. To address this, a ResUNet-based deep learning model is implemented for precise segmentation of brain tumors using the BRISC (Brain Tumor Segmentation and Classification) dataset. The dataset consists of 6,000 contrast-enhanced T1-weighted MRI scans across three major tumor types: glioma, meningioma, and pituitary.

Beyond segmentation, the segmented tumor regions are further classified using a ResNet-based classifier to improve diagnostic utility. Grad-CAM is applied to the segmentation outputs to provide explainability, enabling visual interpretation of regions influencing model predictions. The performance of the proposed ResUNet segmentation model is compared with baseline U-Net and ResNet-based architectures reported in the BRISC benchmarks. Evaluation metrics—including Dice Similarity Coefficient (DSC), Intersection over Union (IoU), and Hausdorff Distance (HD95)—are used to measure segmentation accuracy and boundary precision. The study demonstrates the effectiveness of combining ResUNet segmentation, ResNet classification, and Grad-CAM–based explainability to enhance both performance and interpretability in brain tumor analysis.

Keywords: Brain Tumor Segmentation and Classification, ResUNet, U-Net, ResNet, MRI, BRISC Dataset, Grad-CAM, Explainable AI.

1. Introduction

Magnetic Resonance Imaging (MRI) has become an essential diagnostic tool in medical practice, particularly for the detection and analysis of brain abnormalities. The high-resolution images produced by MRI allow clinicians to examine structural and pathological features of the brain in great detail. However, manually delineating tumor regions remains a complex, time-consuming, and often subjective task. The increasing demand for faster, consistent, and interpretable diagnostic assistance has encouraged researchers to explore Artificial Intelligence (AI) and, more specifically, deep learning approaches for medical image segmentation and classification.

Accurate brain tumor segmentation is crucial for neuro-oncology, yet the process is traditionally labor-intensive and prone to inter-observer variability. Deep Convolutional Neural Networks (DCNNs), such as U-Net, have become standard tools for medical image segmentation; however, their performance can be limited when dealing with diverse tumor shapes, complex boundaries, and deeper architectures where vanishing gradients hinder learning.

To address these challenges, this study proposes a ResUNet-based segmentation model, which integrates residual connections into the U-Net framework to enable deeper feature extraction and improved gradient flow.

In addition to segmentation, the segmented tumor regions are classified into glioma, meningioma, and pituitary tumor categories using a ResNet-based classifier. To strengthen interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to both the segmentation output and classification results, allowing visualization of the regions that most influence the model's decisions. This explainability component is essential in clinical AI systems to build trust and assist radiologists in validating automated predictions.

The proposed pipeline is evaluated using the new BRISC (Brain Tumor Segmentation and Classification) dataset, which provides 6,000 contrast-enhanced T1-weighted MRI scans spanning three major tumor types. Model performance is compared against established U-Net and ResNet baselines using evaluation metrics such as Dice Similarity Coefficient (DSC) and Intersection over Union (IoU). Overall, this study aims to demonstrate the effectiveness of combining ResUNet segmentation, ResNet classification, and Grad-CAM–based explainability for robust and interpretable brain tumor analysis.

2. Motivation

Brain tumors pose a major threat to human health, and early detection is critical for effective treatment planning and improved patient outcomes. Magnetic Resonance Imaging (MRI) is the preferred non-invasive imaging modality for examining brain abnormalities. However, manual segmentation and classification of tumors from MRI scans remain challenging due to several factors:

- **High variability** in tumor size, shape, and location
- **Low contrast** and noise in MRI images
- **Time-consuming and expertise-dependent** manual delineation
- **Inter-observer variability**, where different radiologists may annotate tumors differently

With increasing demand for consistent, automated, and fast diagnostic support, advanced deep learning models have become crucial tools in medical image analysis.

3. Objectives

The main objectives of this work are:

1. **Segmentation:** Implement a ResUNet model to accurately identify and extract brain tumor regions from MRI images.
2. **Classification:** Fine-tune a ResNet18 model to classify the segmented tumor regions into glioma, meningioma, pituitary, or no tumor.
3. **Explainability:** Apply Grad-CAM to visualize and interpret the regions influencing segmentation and classification results.

4. **Evaluation:** Assess system performance using Dice, IoU, accuracy, precision, recall, and F1-score, and validate its effectiveness on the BRISC dataset.

4. Methodology

This study aims to perform accurate and interpretable brain tumor segmentation and classification from MRI scans using a ResUNet-based deep learning framework. The proposed method consists of four main stages: data preprocessing, tumor segmentation using ResUNet, classification of the segmented tumor region using a ResNet model, and explainability analysis using Grad-CAM.

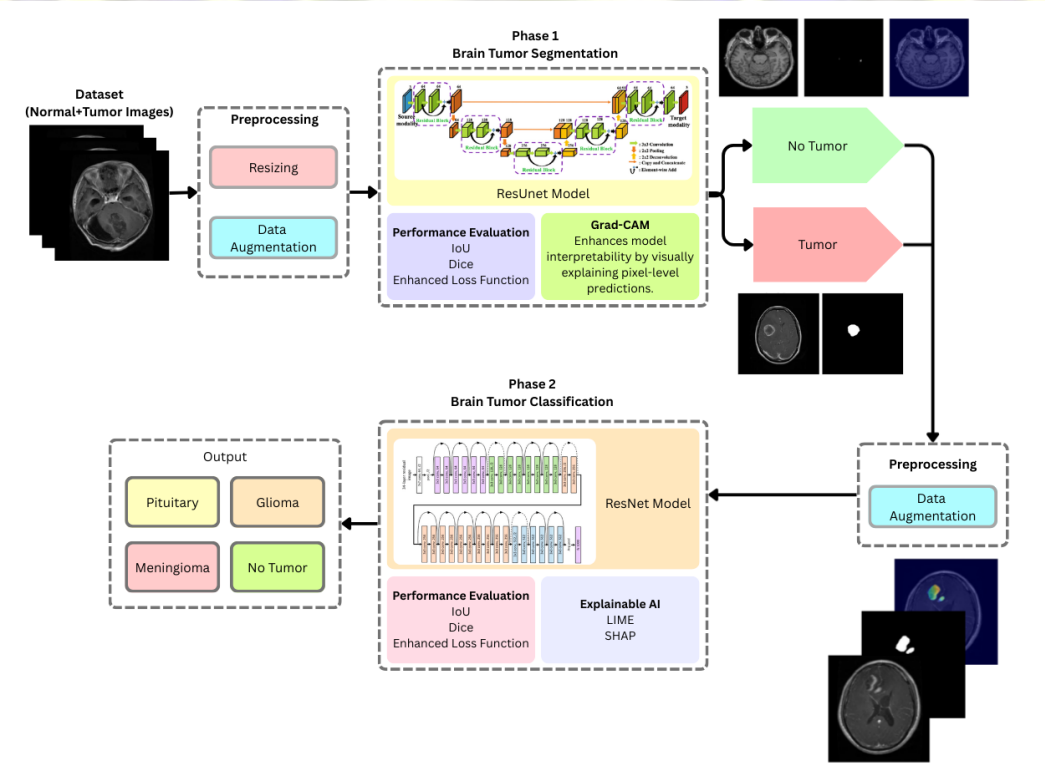


Fig 2.1 - User Workflow of our proposed model

First, raw brain MRI images undergo preprocessing steps such as intensity normalization, resizing, and noise reduction to ensure consistency across the dataset. The preprocessed images are then passed into the trained ResUNet model, which generates a pixel-level segmentation mask highlighting the tumor region. The segmented region is subsequently cropped and extracted from the original MRI scan to isolate the tumor area more accurately.

This cropped tumor region is fed into a trained ResNet classifier, which identifies the tumor type—glioma, meningioma, pituitary, or no tumor. To enhance the interpretability of the entire pipeline, Grad-CAM visualization is applied both to the ResUNet segmentation output and the ResNet classification output. For segmentation, Grad-CAM highlights regions most influential in mask generation, while for classification, it reveals the discriminative areas that drive the model's tumor-

type predictions. This dual-stage explainability provides clinicians with valuable insights and increases trust in the automated system.

The complete workflow integrates segmentation, classification, and explainability, enabling a comprehensive and interpretable brain tumor analysis pipeline.

4.1. Dataset

The BRISC (Brain Tumor Segmentation and Classification) dataset (Fateh et al., 2025) is a large-scale, high-quality MRI dataset designed for brain tumor detection and segmentation.

Feature	Description
Image Type	Contrast-enhanced T1-weighted MRI
Tumor Classes	Glioma, Meningioma, Pituitary
Annotation	Expert-labeled segmentation masks
Imaging Planes	Axial, Coronal, Sagittal
Segmentation Training Set	3933
Segmentation Testing Set	860
Classification Training Set	5000
Classification Testing Set	1000

Table 2.1.1 – BRISC dataset information

This dataset provides both image and mask pairs suitable for supervised segmentation tasks. The original paper also reports baseline results using U-Net and ResNet-based architectures, providing a benchmark for performance comparison.

4.2. Segmentation Model

The segmentation model used in this work is based on a ResUNet architecture, which combines the strengths of the traditional U-Net with residual learning. The model outputs a single-channel segmentation mask that highlights the tumor area in MRI scans and Grad-CAM is also applied to generate the heatmap. By integrating both high-level semantic information and fine spatial details, the ResUNet effectively captures irregular tumor boundaries and provides robust pixel-level predictions, making it well-suited for medical image segmentation tasks.

4.2.1 Data Preprocessing

To train the ResUNet-based brain tumor segmentation model, several preprocessing steps were applied to standardize the BRISC2025 dataset and improve model robustness.

4.2.1.1. Resizing

To maintain consistent input dimensions, all extracted patches and corresponding masks were **resized to 256 × 256 pixels**.

4.2.1.2. Patching

Since the original MRI images are large, each 256 x 256 scan was used as a single patch.

- **Overlap = 0** for training
- Used efficient in-memory patch generation based on our EnhancedPatchDataset class.

Patching reduces GPU memory usage and helps the model focus on local tumor structures.

4.2.1.4. Data Augmentation

Augmentations were used to improve generalization. Random transformations applied include:

- Horizontal flipping
- Vertical flipping
- Rotation
- Elastic deformation
- Gaussian noise
- Random brightness and contrast
- Z-Score Normalization

These augmentations increased diversity and prevented overfitting.

4.2.2. Model Architecture

The segmentation model is built using the **ResUNet** architecture, combining U-Net's encoder–decoder design with **ResNet-style residual blocks** for deeper feature extraction.

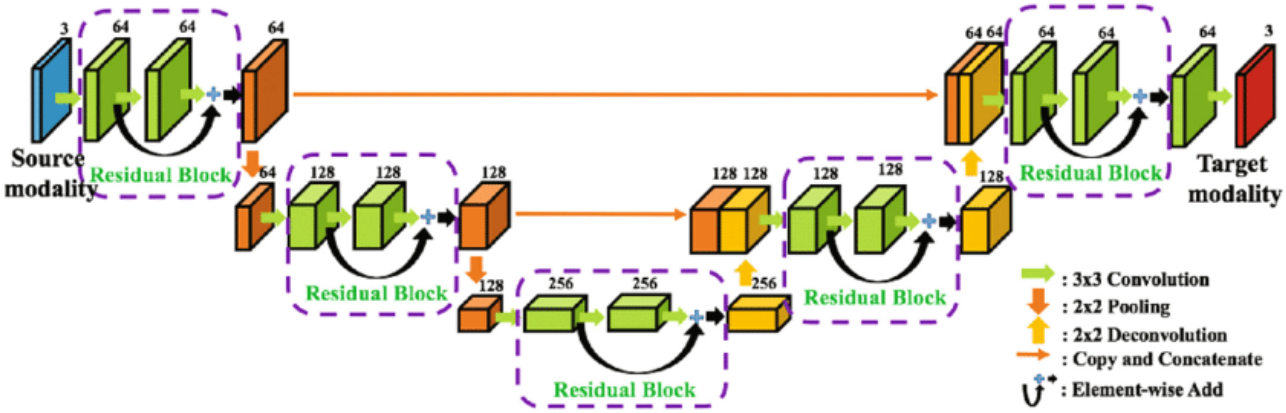


Fig 2.2.2.1 – ResUNet architecture

Source: http://www.researchgate.net/figure/llustration-of-the-proposed-Res-Unet-architecture-as-the-generator_fig2_327748708

Encoder

- Consists of 4 **ConvBlock** layers
- Each block includes:
  - Two convolution layers
  - Batch Normalization
  - ReLU activation
  - A residual shortcut connection

This design enhances gradient flow and reduces vanishing gradients.

Bottleneck

The bottleneck captures high-level semantic features using a **512-channel ConvBlock**.

Decoder

The decoder mirrors the encoder using:

- **Transposed convolutions** for upsampling
- **ConvBlockDec** modules
- **Skip connections** connecting encoder and decoder features

This helps reconstruct fine tumor boundaries.

Residual Block

A Residual Block in ResUNet is the core component that uses skip connections (shortcut connections) to add the original input to the output of a few convolutional layers, allowing the network to learn "residual functions" (the difference) rather than the full mapping, which helps train

very deep networks by preventing vanishing gradients and overcoming the degradation problem, making deep models effective for tasks like image segmentation.

Copy and Concatenate Process

Copy and concatenate operations are fundamental to the network's U-shaped design, serving to merge high-resolution spatial information from the **encoder path** (contracting path) with the upsampled features in the **decoder path** (expansive path). This helps in accurate pixel-level localization for tasks like image segmentation.

Final Layer

A **1×1 convolution** maps the decoder output to a single-channel segmentation mask.

Key Architectural Advantages

- Robust gradient propagation through residual paths
- Better localization through skip connections
- Supports Grad-CAM visualization on the final convolution layer

Layer (type)	Output Shape	Param #
Conv2d-1	[-1, 64, 256, 256]	576
Conv2d-2	[-1, 64, 256, 256]	64
BatchNorm2d-3	[-1, 64, 256, 256]	128
ReLU-4	[-1, 64, 256, 256]	0
Conv2d-5	[-1, 64, 256, 256]	36,864
ConvBlock1-6	[-1, 64, 256, 256]	0
BatchNorm2d-7	[-1, 64, 256, 256]	128
ReLU-8	[-1, 64, 256, 256]	0
Conv2d-9	[-1, 128, 128, 128]	8,192
Conv2d-10	[-1, 128, 128, 128]	73,728
BatchNorm2d-11	[-1, 128, 128, 128]	256
ReLU-12	[-1, 128, 128, 128]	0
Conv2d-13	[-1, 128, 128, 128]	147,456
ConvBlock-14	[-1, 128, 128, 128]	0
BatchNorm2d-15	[-1, 128, 128, 128]	256
ReLU-16	[-1, 128, 128, 128]	0
Conv2d-17	[-1, 256, 64, 64]	32,768
Conv2d-18	[-1, 256, 64, 64]	294,912
BatchNorm2d-19	[-1, 256, 64, 64]	512
ReLU-20	[-1, 256, 64, 64]	0
Conv2d-21	[-1, 256, 64, 64]	589,824
ConvBlock-22	[-1, 256, 64, 64]	0
BatchNorm2d-23	[-1, 256, 64, 64]	512
ReLU-24	[-1, 256, 64, 64]	0
Conv2d-25	[-1, 512, 32, 32]	131,072
Conv2d-26	[-1, 512, 32, 32]	1,179,648
BatchNorm2d-27	[-1, 512, 32, 32]	1,024
ReLU-28	[-1, 512, 32, 32]	0
Conv2d-29	[-1, 512, 32, 32]	2,359,296
ConvBlock-30	[-1, 512, 32, 32]	0
BatchNorm2d-31	[-1, 512, 32, 32]	1,024
ReLU-32	[-1, 512, 32, 32]	0
Conv2d-33	[-1, 1024, 16, 16]	524,288
Conv2d-34	[-1, 1024, 16, 16]	4,718,592
BatchNorm2d-35	[-1, 1024, 16, 16]	2,048
ReLU-36	[-1, 1024, 16, 16]	0
Conv2d-37	[-1, 1024, 16, 16]	9,437,184
ConvBlock-38	[-1, 1024, 16, 16]	0
ConvTranspose2d-39	[-1, 512, 32, 32]	2,097,664
BatchNorm2d-40	[-1, 1024, 32, 32]	2,048
ReLU-41	[-1, 1024, 32, 32]	0
Conv2d-42	[-1, 512, 32, 32]	524,288
Conv2d-43	[-1, 512, 32, 32]	4,718,592
BatchNorm2d-44	[-1, 512, 32, 32]	1,024
ReLU-45	[-1, 512, 32, 32]	0
Conv2d-46	[-1, 512, 32, 32]	2,359,296
ConvBlock-47	[-1, 512, 32, 32]	0
ConvBlockDec-48	[-1, 512, 32, 32]	0
ConvTranspose2d-49	[-1, 256, 64, 64]	524,544
BatchNorm2d-50	[-1, 512, 64, 64]	1,024
ReLU-51	[-1, 512, 64, 64]	0
Conv2d-52	[-1, 256, 64, 64]	131,072
Conv2d-53	[-1, 256, 64, 64]	1,179,648
BatchNorm2d-54	[-1, 256, 64, 64]	512
ReLU-55	[-1, 256, 64, 64]	0
Conv2d-56	[-1, 256, 64, 64]	589,824
ConvBlock-57	[-1, 256, 64, 64]	0
ConvBlockDec-58	[-1, 256, 64, 64]	0
ConvTranspose2d-59	[-1, 128, 128, 128]	131,200
BatchNorm2d-60	[-1, 256, 128, 128]	512
ReLU-61	[-1, 256, 128, 128]	0
Conv2d-62	[-1, 128, 128, 128]	32,768
Conv2d-63	[-1, 128, 128, 128]	294,912
BatchNorm2d-64	[-1, 128, 128, 128]	256
ReLU-65	[-1, 128, 128, 128]	0
Conv2d-66	[-1, 128, 128, 128]	147,456
ConvBlock-67	[-1, 128, 128, 128]	0
ConvBlockDec-68	[-1, 128, 128, 128]	0
ConvTranspose2d-69	[-1, 64, 256, 256]	32,832
BatchNorm2d-70	[-1, 128, 256, 256]	256
ReLU-71	[-1, 128, 256, 256]	0
Conv2d-72	[-1, 64, 256, 256]	8,192
Conv2d-73	[-1, 64, 256, 256]	73,728
BatchNorm2d-74	[-1, 64, 256, 256]	128
ReLU-75	[-1, 64, 256, 256]	0
Conv2d-76	[-1, 64, 256, 256]	36,864
ConvBlock-77	[-1, 64, 256, 256]	0
ConvBlockDec-78	[-1, 64, 256, 256]	0
Conv2d-79	[-1, 1, 256, 256]	64

Total params: 32,429,056
Trainable params: 32,429,056
Non-trainable params: 0

Input size (MB): 0.25
Forward/backward pass size (MB): 1212.50
Params size (MB): 123.71
Estimated Total Size (MB): 1336.46

Fig 2.2.2.2 – ResUNet Parameters

4.2.3. Enhanced Loss Function

To handle class imbalance and improve both pixel-wise accuracy and boundary precision, an **Enhanced Hybrid Loss** was used combining:

1. Binary Cross-Entropy (BCE) Loss

Measures pixel-level classification accuracy.

2. Dice Loss

Captures region-level overlap between prediction and ground truth.

Final Loss Function

Binary Cross Entropy Loss is given by:

$$\boldsymbol{\mathcal{L}_{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}[y_i.log(\sigma(z_i)) + (1-y_i).log(1 - \sigma(z_i))]$$

N— total number of pixels in the mask, $y_i \in \{0,1\}$— ground truth label for pixel i,

z_i— raw logit (model output before sigmoid),

$\sigma(z_i) = \frac{1}{1+e^{-z_i}}$— sigmoid function → converts logits into probabilities between 0 and 1

In PyTorch, this is efficiently implemented using:

nn.functional.binary_cross_entropy_with_logits(logits, target)

This component ensures that the model learns pixel-level distinctions between tumor and non-tumor tissues.

The Dice coefficient is defined as:

$$\boldsymbol{Dice} = \frac{2\sum_{i=1}^{N} p_i\, y_i + \epsilon}{\sum_{i=1}^{N} p_i^2 + \sum_{i=1}^{N} y_i^2 + \epsilon}$$

$p_i = \sigma(z_i)$— predicted probability for pixel i, $y_i \in \{0,1\}$— true label,

ϵ— very small number (like 10^{-6}) to avoid division by zero

The Dice Loss is then expressed as:

$$\boldsymbol{\mathcal{L}_{Dice}} = 1 - Dice$$

The final loss function is defined as a weighted combination of both components.

$$\boldsymbol{\mathcal{L}_{final}} = 0.4 \cdot \mathcal{L}_{BCE} + 0.6 \cdot \mathcal{L}_{Dice}$$

This term encourages the model to maximize the overlap between the predicted tumor mask and the true tumor region, thereby improving segmentation quality, especially around complex or diffuse tumor boundaries.

The network parameters W are optimized using the Adamw optimizer with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-5}. The integration of BCE and Dice components allows the model to achieve both pixel-wise classification accuracy and region-level consistency, resulting in smoother, more accurate, and clinically meaningful segmentation of brain tumor regions compared to traditional loss functions such as Mean Squared Error (MSE) or BCE alone.

Optimizations:

- **Optimizer:** AdamW
- **Learning rate:** 1×10^{-4}
- **Weight decay:** 1×10^{-5}
- **Scheduler:** ReduceLROnPlateau

This hybrid formulation improves segmentation accuracy, especially around irregular tumor boundaries.

4.2.4. Evaluation Metrics

Model performance was evaluated using standard medical image segmentation metrics:

1. Dice Similarity Coefficient (DSC)

Measures the overlap between predicted and ground truth masks.

$$DSC = \frac{2 \mid P \cap G \mid}{\mid P \mid + \mid G \mid}$$

P = Predicted Mask
G = Ground Truth Mask

2. Intersection over Union (IoU)

Evaluates the ratio between intersection and union of predicted and ground truth masks.

$$IoU = \frac{\mid P \cap G \mid}{\mid P \cup G \mid}$$

P = Predicted Mask
G = Ground Truth Mask

4.2.5. Grad-CAM Explainability

To enhance the interpretability of the ResUNet segmentation model, **Gradient-Weighted Class Activation Mapping (Grad-CAM)** was applied to visualize the regions of the MRI scan that most strongly influenced the model's segmentation predictions. Explainability is particularly important in medical deep learning applications, where clinicians must trust and understand how the model arrives at its decisions.

4.2.6. Result

Using the optimized ResUNet model, the following results were achieved:

- **Dice Score:** 0.8423
- **IoU:** 0.7664

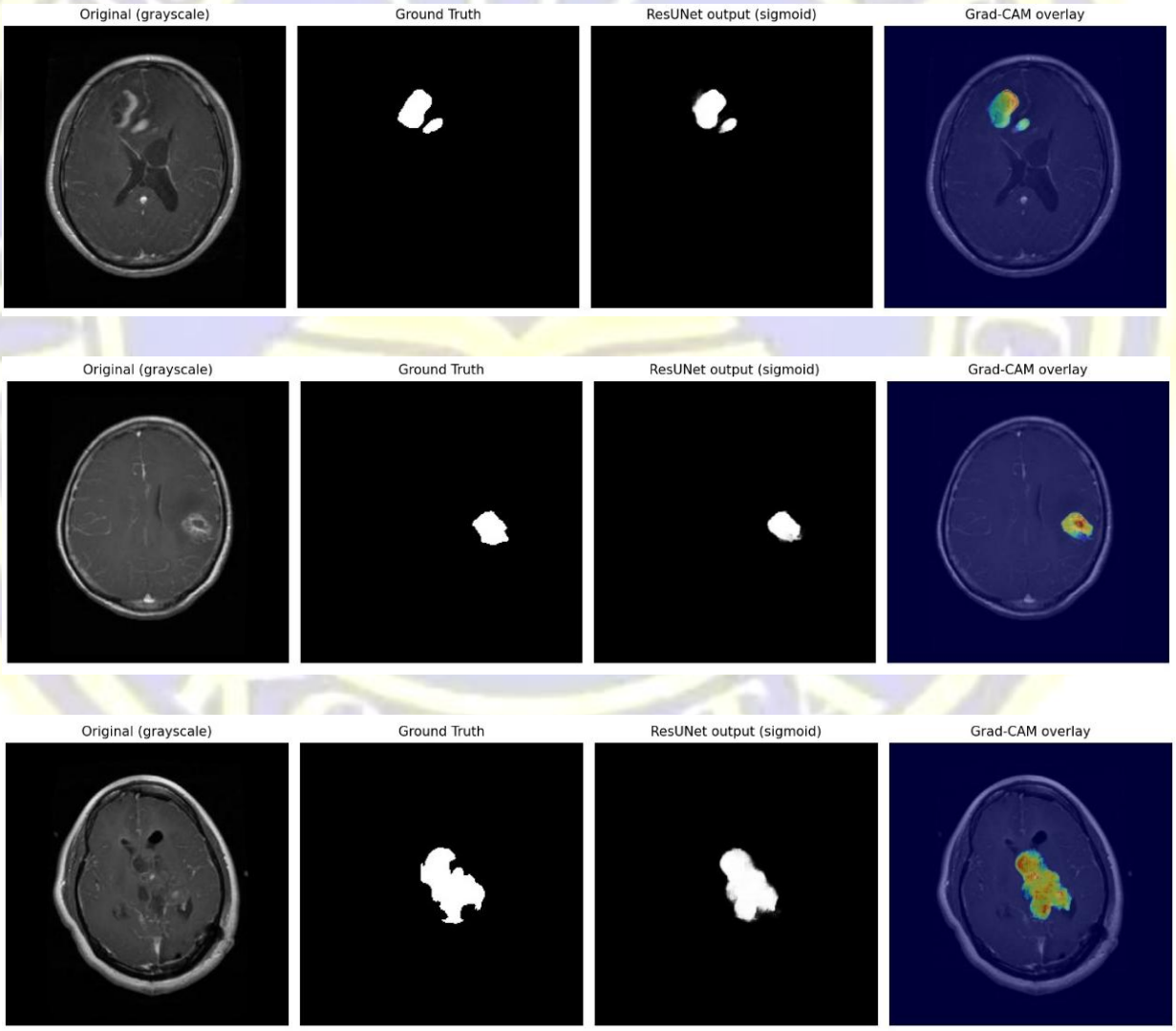


Fig 2.2.6.1 – Segmentation Model outputs (Segmentation Mask + Grad-CAM Heatmap)

4.2.7. Observations

- Segmentation masks closely match tumor shapes
- Edges are smooth due to residual learning
- Works well on glioma, meningioma, and pituitary tumor variations

4.3. Purpose of Grad-CAM

Grad-CAM provides heatmaps that highlight the spatial regions contributing to the network's output. In the context of tumor segmentation, these maps help in:

- Visualizing which areas guided the model during tumor mask generation
- Ensuring the model is focusing on actual pathological structures, not irrelevant artifacts
- Supporting radiologists and clinicians in understanding model predictions
- Validating that the model's feature learning aligns with clinical expectations

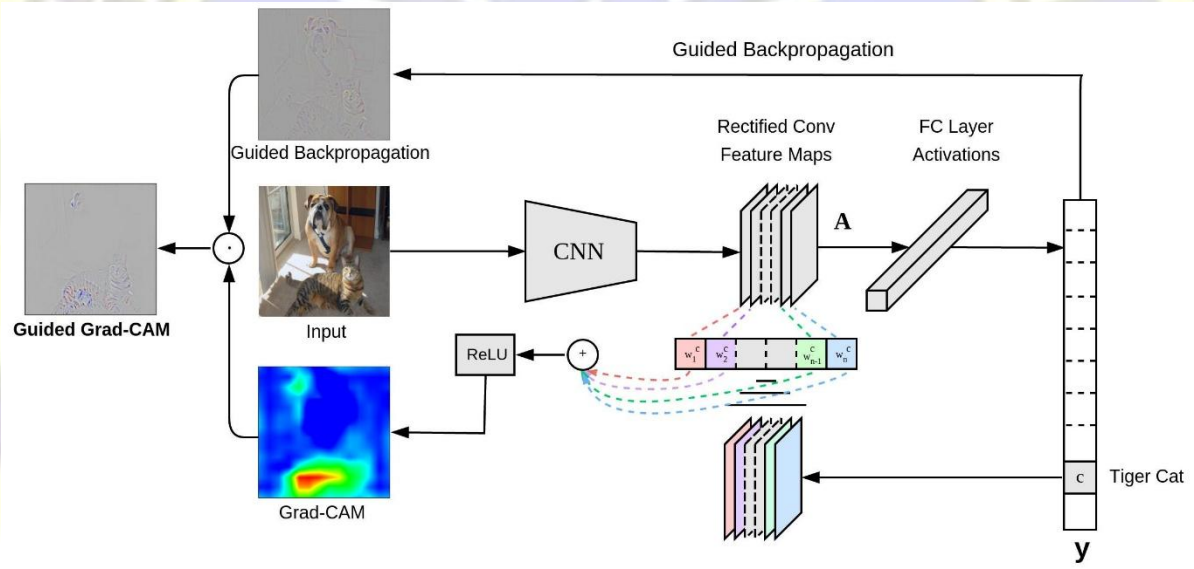


Fig 2.7.1.1 – Grad-CAM architecture

Source: https://miro.medium.com/v2/format:webp/1*hHPn81BbKEl7xDsHr5aSIA.png

4.3.1. Grad-CAM Implementation

Grad-CAM implementation was integrated into the ResUNet pipeline based on the final convolution layer.
This layer was selected because:

- It retains high-level semantic information

- It directly influences the final segmentation output
- It allows visualization of precise activation regions at the mask level

The Grad-CAM module computes:

1. **Forward activations** from the target convolution layer
2. **Backward gradients** flowing from the model output
3. **Channel-wise importance weights** obtained by global average pooling of gradients
4. A **class-discriminative heatmap** that highlights influential pixels

4.3.2. Visualization of Model Attention

For each test MRI scan, the following were generated:

- Original MRI image (grayscale)
- Ground-truth tumor mask
- ResUNet output mask after sigmoid activation
- Grad-CAM heatmap overlay

These overlays show that the ResUNet model consistently focuses on:

- Core tumor regions
- Tumor boundaries
- Contrast-enhanced areas associated with abnormal tissue growth

4.4. Classification Model

After segmenting the tumor region using the ResUNet model, the extracted tumor area is used as input to a fine-tuned ResNet18 classifier. The goal of the classifier is to categorize the tumor into one of the three classes—**glioma, meningioma, pituitary**—or **no tumor**. This two-stage pipeline (segmentation → classification) improves diagnostic precision by ensuring the classifier focuses only on the tumor region.

4.4.1. Data Preprocessing

The classification pipeline utilized three different sources of information for each input sample:
1. The original grayscale image,
2. The segmentation mask generated by the ResUNet model, and
3. The Grad-CAM heatmap computed from the segmentation output.

4.4.1.1. Resizing Images

- All images are resized to a fixed input size (e.g., 256 x 256 or model-specific size).
- Ensures compatibility with the ResNet-based classifier.

4.4.1.2. Channel Conversion to 3-Channel Format

- The resized grayscale image was then forwarded to the ResUNet model, which produced a binary segmentation mask corresponding to the region of interest along with the Grad-CAM heatmap.
- Both the mask and Grad-CAM output were normalized to the range [0, 1].
- The three elements are then stacked along the channel dimension to form a 3-channel tensor:

$$X = [I_{\text{gray}}, M_{\text{seg}}, H_{\text{cam}}]$$

I_{gray} = Original Resized Image, M_{seg} = Segmentation Mask for the Resized Image,
H_{cam}= Grad-CAM Heatmap

- This 3-channel fused representation was used as the input to the classification network. Standard normalization was applied to the image channel, while the mask and Grad-CAM channels were left in normalized scalar form, ensuring stable training.

4.4.1.3. Label Encoding

- Class labels are converted into numerical form (e.g., 0 for Glioma, 1 for Meningioma and 2 for No Tumor and 3 for Pituitary).

4.4.1.4. Train–Validation–Test Split

- Dataset is split to evaluate the model fairly and prevent data leakage.
- 12% of the Training dataset has been used for validation purpose.
- The rest has been used for training the classification model(ResNet18).

4.4.2. Data Augmentation

- Horizontal Flip
- Vertical Flip
- Random Rotate
- ShiftScaleRotate:
    - **Shift:** Moves the image slightly (up to 5% of image size).
    - **Scale:** Zooms in/out up to 10% to simulate distance variation.
    - **Rotate:** Rotates the image within ±15°.
- Normalization

4.4.3. Model Architecture

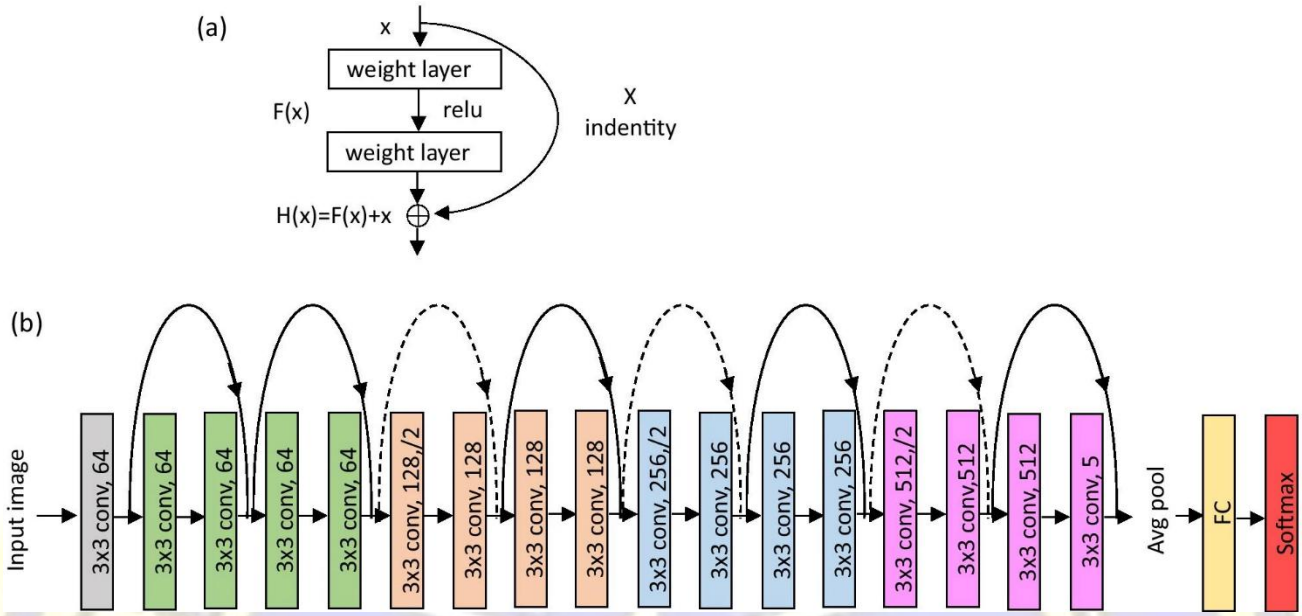


Fig 2.3.2.1 – (a) Residual Learning Block, (b) ResNet18 architecture

Source: https://www.researchgate.net/figure/a-The-basic-residual-block-b-The-ResNet18-architecture-The-numbers-in-each_fig1_352666166

The classification model is based on a **custom implementation of ResNet-18**, consisting of:

Initial Convolution Stage

- A **7×7 convolution** with stride 2
- Batch normalization
- ReLU activation
- **3×3 max-pooling**

This stage extracts low-level textures from the input MRI crop.

Residual Learning Blocks

The architecture follows the standard residual structure:

a. Identity Shortcut Block

Used when input and output dimensions match:

- Two 3×3 convolutions
- BatchNorm + ReLU
- Residual addition: $y = F(x) + x$

b. Projection Shortcut Block

Used when spatial dimensions change:

- First convolution uses stride 2 for downsampling
- A 1×1 projection convolution aligns dimensions
- Residual addition ensures stable gradient flow

Layer Configuration

- **Layer 1:** 2 × Identity Blocks (64 channels)
- **Layer 2:** Projection Block (64→128), then Identity Block
- **Layer 3:** Projection Block (128→256), then Identity Block
- **Layer 4:** Projection Block (256→512), then Identity Block

This hierarchical design extracts increasingly abstract features across depth levels.

Classification Head

- **Adaptive Average Pooling** to 1×1
- **Fully connected layer** (512 → 4 classes)
- Outputs logits for softmax classification

Weight Initialization

Our code applies **Kaiming Normal initialization**. This improves convergence when using ReLU activations.

4.4.4. Loss Function

A **cross-entropy loss** function was employed to optimize the classification network, as it is well-suited for multi-class and binary classification tasks. The loss is defined as:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

C is the number of classes, y_i is the ground truth label (one-hot encoded),

$\hat{y}_i$ is the predicted probability produced by the softmax output layer.

Cross-entropy loss penalizes incorrect predictions proportionally to their confidence, promoting stable and convergent training. Additionally, the optimizer used (such as Adam or SGD with momentum) ensured efficient gradient updates during back-propagation.

4.4.5. Evaluation Metrics

The performance of the classifier was evaluated using the following metrics:

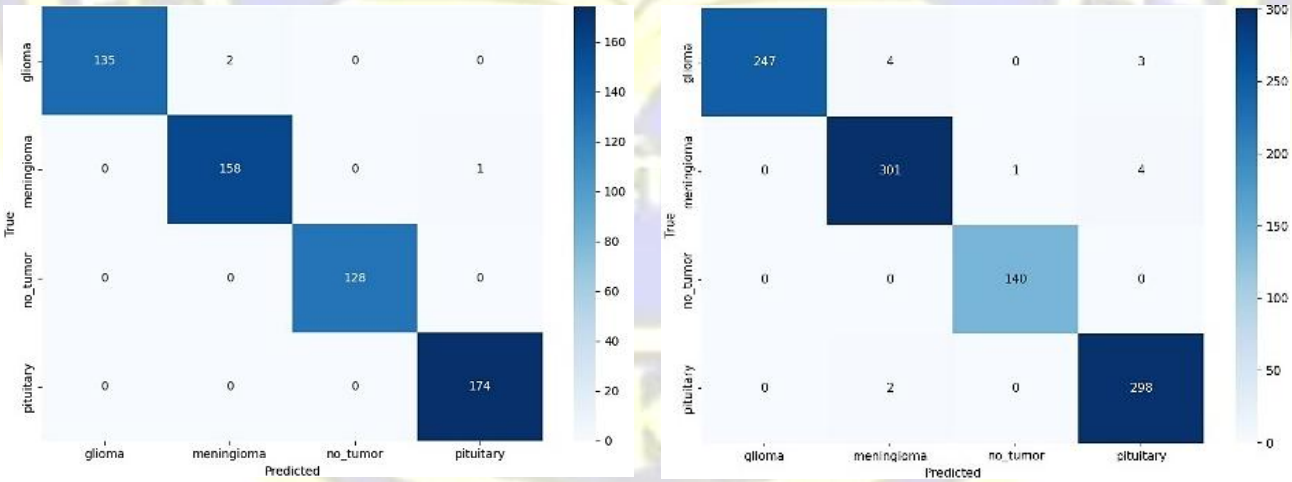
- **Accuracy** – measures the proportion of correctly classified samples.
- **Precision** – assesses how many predicted positives are true positives.
- **Recall** – measures the model's ability to correctly identify positive samples.
- **F1-Score** – harmonic mean of precision and recall, especially useful for imbalanced datasets.
- **Confusion Matrix** – provides detailed insights into class-wise performance and misclassification patterns.

These metrics offered a comprehensive understanding of model robustness, particularly in cases where datasets exhibit class imbalance or when false positives/negatives carry different clinical or practical significance.

4.4.6. Result

For our classification model:

- Training Accuracy: 99.08%
- Test Accuracy: 98.60%



(a) (b)

Fig 2.4.6.1 – Confusion Matrix of (a) Training and (b) Testing dataset

5. Future Work

In the next phase of this research, we plan to significantly extend and enhance the current segmentation and classification pipeline. Future work will focus on improving model performance, incorporating stronger architectures, and increasing the system's usability and interpretability.

To enhance segmentation accuracy, we aim to integrate more advanced backbone models such as nnU-Net or SwinUNet (Swin Transformer-based U-Net). These architectures offer improved adaptability, stronger feature representation, and state-of-the-art performance on a wide range of medical imaging tasks. By adopting these models, we expect to achieve more precise tumor boundaries and better generalization across diverse datasets.

On the classification side, we will explore upgrading from the current ResNet-based classifier to more powerful transformer-based or hybrid architectures if they provide measurable accuracy improvements. This will help improve diagnostic reliability and reduce misclassification.

Interpretability will also be a major component of future development. Beyond Grad-CAM, we plan to apply LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive exPlanations) to provide deeper, feature-level explanations for both segmentation and classification outputs. These XAI tools will help clinicians understand why the model makes certain predictions, increasing trust and transparency in the system.

Finally, we intend to develop a web-based user interface that allows clinicians to upload MRI scans, view segmentation and classification results, and interactively explore explainability visualizations generated by Grad-CAM, LIME, and SHAP. This interface will bridge the gap between deep learning research and real-world clinical workflows, making the system accessible, interpretable, and practical for medical decision support.

6. Literature Survey

Explainable Artificial Intelligence (XAI) has become increasingly important in medical imaging to make deep learning (DL) models more transparent and trustworthy. In this field, XAI techniques are mainly divided into perturbation-based and gradient-based methods. Perturbation-based approaches (such as LIME, SHAP, occlusion, and deconvolution) study the model by slightly changing the input image and checking how predictions change. Although these methods are intuitive, they are slow and computationally heavy for 3D MRI data.

Gradient-based approaches (such as Grad-CAM, Guided Backpropagation, Integrated Gradients, and SmoothGrad) use the model's internal gradients to highlight which image regions influenced its decision. These are faster, post-hoc techniques that can be applied without retraining the model.

Some researchers have applied XAI to tumor segmentation tasks. While 3D versions of Grad-CAM and CAM have been explored, they often increase model complexity or reduce accuracy.

Frameworks such as NeuroXAI have recently combined multiple XAI methods to produce 2D and 3D explanations for MRI-based brain tumor analysis without changing model performance.

Despite progress, current studies still have limitations: most explanations are qualitative (visual heatmaps), rely on 2D slices, and rarely include clinical validation or feedback from radiologists. There is also little research on how explanations behave across different scanners or datasets. Therefore, there is a strong need for quantitative, 3D, and clinically reliable XAI methods for brain tumor detection.

Identified Research Gaps:

Gap A: No standard quantitative way to evaluate XAI results — heatmaps are subjective.

Gap B: Lack of 3D explainability — most studies work only on 2D slices.

Gap C: XAI is used after training; not integrated into model learning.

Gap D: Few human-in-the-loop or radiologist validation studies.

Gap E: Explanations are not robust across scanners or datasets.

Gap F: No uncertainty or confidence shown with explanations.

Gap G: Limited XAI work on tumor grading or prognosis prediction.

Gap H: No lightweight or real-time XAI systems for clinical use.

7. Conclusion

This project successfully implements an end-to-end deep learning framework for automatic brain tumor segmentation and classification using MRI images from the BRISC dataset. By integrating ResUNet for segmentation, ResNet18 for classification, and Grad-CAM for model interpretability, the proposed system delivers both high accuracy and clinical transparency.

The ResUNet model achieved strong performance in identifying tumor regions, producing smooth and well-localized segmentation masks with competitive Dice and IoU scores. The fine-tuned ResNet18 classifier, operating on precisely cropped tumor regions, effectively distinguished between glioma, meningioma, pituitary tumors, and non-tumor cases. This two-stage hierarchical pipeline ensures that classification decisions are based exclusively on tumor-specific features rather than surrounding brain structures.

Furthermore, the inclusion of Grad-CAM explainability visualizations provides valuable insights into the internal decision-making process of the models, reinforcing the trust and reliability of the system. The ability to visualize activation maps makes the approach highly applicable for real clinical environments where transparency is essential.

Overall, the study demonstrates that combining segmentation and classification in a unified workflow significantly enhances diagnostic capability and reduces the subjectivity associated with

manual interpretation. The results highlight the potential of deep learning systems to assist radiologists by providing fast, accurate, and interpretable brain tumor analysis.

8. References

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